

Two Conditional Reductions for the Corson Compact $\ell_1(\omega_2)$ Problem

arXiv automatic math run `fa_banach_001`

Candidate conditional packet, July 3, 2026

Classification

- **Status:** conditional result; not a full solution of the source consistency question.
- **Source:** P. Hajek and T. Russo, *An uncountable version of Ptak's combinatorial lemma*, arXiv:1805.06028; J. Math. Anal. Appl. 470 (2019), 1070–1080.
- **Target:** Problem 4.3 asks whether the existence of a Corson compact K such that $\ell_1(\omega_2) \hookrightarrow C(K)$ is consistent with ZFC.
- **Support paper:** W. Marciszewski, G. Plebanek, and K. Zakrzewski, *Digging into the classes of κ -Corson compact spaces*, arXiv:2107.02513.

Source Question

On page 11, Problem 4.3 of the source asks:

Is the existence of a Corson compact K such that $\ell_1(\omega_2) \hookrightarrow C(K)$ consistent with ZFC?

The source notes immediately afterward that the answer is negative under CH and also under MA_{ω_1} .

In accordance with Argyros' results on Pelczyński's conjecture that we mentioned above, it is natural to conjecture that Haydon's equivalence may actually be valid for every cardinal number $\kappa \geq \omega_2$. This would, of course, imply the validity of Pták's lemma for every $\kappa \geq \omega_2$.

In case that the conjecture were true, it would also lead to a negative answer to the following question.

Problem 4.3. Is the existence of a Corson compact K such that $\ell_1(\omega_2) \hookrightarrow C(K)$ consistent with ZFC?

Let us just note that, under CH, such a compact space can not exist, in light of Theorem 4.2 and the fact that continuous images of Corson compact are Corson compact ([14, 27]), while $[0, 1]^{\omega_2}$ is not Corson. Such a compact space also fails to exist under MA_{ω_1} , according to the results we recorded at the beginning of Section 3.

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Conditional Dependencies

This packet isolates two independent set-theoretic hypotheses.

Assumption 1 (Measure caliber at ω_2). ω_2 is a caliber of Radon measures: for every compact space L , every $\mu \in P(L)$, and every family $(A_\xi)_{\xi < \omega_2} \subseteq \text{Bor}(L)$ of μ -positive sets, there is $x \in L$ such that

$$|\{\xi < \omega_2 : x \in A_\xi\}| = \omega_2.$$

Assumption 2 (Point-countable positive family at ω_2). There are a compact Hausdorff space L , a Radon probability measure μ on L , and closed sets $C_\xi \subseteq L$, $\xi < \omega_2$, such that $\mu(C_\xi) > 0$ for every $\xi < \omega_2$, and

$$|\{\xi < \omega_2 : x \in C_\xi\}| \leq \omega \quad \text{for every } x \in L.$$

The first hypothesis gives a negative answer. The second gives a positive example. The packet does not prove either hypothesis in ZFC, nor does it claim that they exhaust all possible models.

Proof Intuition

The negative direction uses Talagrand's dual-ball characterization of large $\ell_1(\kappa)$ subspaces. If $\ell_1(\omega_2)$ embeds into $C(K)$, then the weak-star compact dual ball of $C(K)$ maps continuously onto the cube $[0, 1]^{\omega_2}$. For a Corson K , the space K is also ω_2 -Corson. Under the measure-caliber hypothesis, the later κ -Corson theory says that $M_1(K) = B_{C(K)^*}$ is ω_2 -Corson. Regular κ -Corson compacta are preserved by continuous images, while $[0, 1]^{\omega_2}$ is not ω_2 -Corson. This contradiction rules out the desired K .

The positive direction reverses the obstruction. A point-countable family of positive-measure closed sets produces an adequate compact

$$\mathcal{A} = \{A \subseteq \omega_2 : \bigcap_{\xi \in A} C_\xi \neq \emptyset\}.$$

Point-countability makes $\mathcal{A}$ Corson. Positive measure supplies a uniform lower bound on a subfamily of size ω_2 , and hence the Ptak convex mean condition for the finite members of $\mathcal{A}$. The source proposition then embeds $\ell_1(\omega_2)$ into $C(\mathcal{A})$.

Conditional Negative Result

Theorem 1. Assume that ω_2 is a caliber of Radon measures. Then there is no Corson compact K such that $\ell_1(\omega_2) \hookrightarrow C(K)$.

Proof. Let K be Corson compact. Then K is ω_2 -Corson, because countable supports have cardinal $< \omega_2$.

By Theorem 12.4 of Marciszewski–Plebanek–Zakrzewski, for a regular uncountable cardinal κ , the following are equivalent: κ is a caliber of Radon measures; $P(L)$ is κ -Corson for every κ -Corson compact L ; and $M_1(L)$ is κ -Corson for every κ -Corson compact L . Applying this with $\kappa = \omega_2$, the weak-star compact unit ball

$$B_{C(K)^*} = M_1(K)$$

is ω_2 -Corson.

Suppose, toward a contradiction, that $\ell_1(\omega_2) \hookrightarrow C(K)$. Talagrand's theorem, as quoted in the source paper before Theorem 4.2, says that for cardinals κ with $\text{cf}(\kappa) \geq \omega_1$, $\ell_1(\kappa) \hookrightarrow X$ if and only if there is a continuous surjection

$$(B_{X^*}, w^*) \longrightarrow [0, 1]^\kappa.$$

Thus there is a continuous surjection

$$M_1(K) \longrightarrow [0, 1]^{\omega_2}.$$

Theorem 9.2 of Marciszewski–Plebanek–Zakrzewski states that, for regular κ , continuous images of κ -Corson compacta are κ -Corson. Hence $[0, 1]^{\omega_2}$ would be ω_2 -Corson. This contradicts their Corollary 7.2, which says that $[0, 1]^\kappa$ is not κ -Corson for any $\kappa \geq \omega$. $\square$

Conditional Positive Construction

Theorem 2. *Assume there is a compact Radon probability space (L, μ) with a point-countable family of positive-measure closed sets $(C_\xi)_{\xi < \omega_2}$. Then there is a Corson compact space K such that $\ell_1(\omega_2) \hookrightarrow C(K)$.*

Proof. For $A \subseteq \omega_2$, write

$$A \in \mathcal{A} \iff \bigcap_{\xi \in A} C_\xi \neq \emptyset.$$

Equivalently, by compactness of L , $A \in \mathcal{A}$ if and only if every finite subset of A has non-empty C_ξ -intersection. Therefore $\mathcal{A}$ is an adequate compact family, hence a compact subspace of 2^{ω_2} .

The family is hereditary: if $B \subseteq A \in \mathcal{A}$, then $B \in \mathcal{A}$. It is also Corson. Indeed, if $A \in \mathcal{A}$, choose $x \in \bigcap_{\xi \in A} C_\xi$. Then

$$A \subseteq \{\xi < \omega_2 : x \in C_\xi\},$$

and the latter set is countable by the point-countability assumption. Thus every point of $\mathcal{A} \subseteq 2^{\omega_2}$ has countable support.

Since ω_2 is regular and every $\mu(C_\xi)$ is positive, there are $S \subseteq \omega_2$ with $|S| = \omega_2$ and a number $\delta > 0$ such that $\mu(C_\xi) > \delta$ for all $\xi \in S$. Let

$$\mathcal{F} = \{F \in [S]^{<\omega} : \bigcap_{\xi \in F} C_\xi \neq \emptyset\}.$$

This is a hereditary family of finite subsets of S , and its closure in 2^S is the adequate compact

$$K = \{A \subseteq S : \bigcap_{\xi \in A} C_\xi \neq \emptyset\},$$

which is a closed subspace of $\mathcal{A}$, hence Corson.

We verify the Ptak convex mean lower bound from the source proposition. Let λ be any convex mean on S , so λ has finite support and $\sum_{\xi \in S} \lambda(\xi) = 1$. Then

$$\int_L \sum_{\xi \in S} \lambda(\xi) \mathbf{1}_{C_\xi} d\mu = \sum_{\xi \in S} \lambda(\xi) \mu(C_\xi) > \delta.$$

Hence there is $x \in L$ such that

$$\sum_{\xi \in S} \lambda(\xi) \mathbf{1}_{C_\xi}(x) > \delta.$$

Let

$$F_x = \{\xi \in \text{supp}(\lambda) : x \in C_\xi\}.$$

Then F_x is finite, belongs to $\mathcal{F}$, and $\lambda(F_x) > \delta$. Therefore

$$\inf_{\lambda} \sup_{F \in \mathcal{F}} \lambda(F) \geq \delta,$$

where the infimum ranges over convex means on S .

Proposition 2.1 of Hajek–Russo says that this condition implies $\ell_1(S) \hookrightarrow C(\overline{\mathcal{F}}) = C(K)$. Since $|S| = \omega_2$, this is the desired embedding $\ell_1(\omega_2) \hookrightarrow C(K)$. $\square$

Consequences

Corollary 1. *If the point-countable positive family assumption at ω_2 is consistent, then the source Problem 4.3 has a positive consistency answer.*

Proof. Apply Theorem 2 in a model satisfying the assumption. $\square$

Corollary 2. *In every model in which ω_2 is a caliber of Radon measures, the source Problem 4.3 has a negative answer.*

Proof. This is Theorem 1. $\square$

Verification and Novelty Notes

The run registry, solution, attempt, and proof-gap indexes were searched for arXiv:1805.06028, Ptak/Ptak, Corson compact, $\ell_1(\omega_2)$, property (M), Haydon, and the exact open-problem phrase. No exact packet for Problem 4.3 was found. Nearby packets concern WLD/property-(M) and other nonseparable Corson or ω_2 frontiers, but not this problem.

Focused web searches for exact phrases such as “Corson compact $\ell_1(\omega_2)$ $C(K)$ ”, “Haydon ℓ_1 omega_2 Corson compact”, and “caliber of Radon measures omega_2” found the source paper and the later κ -Corson papers, but no paper explicitly answering Problem 4.3. The Marciszewski–Plebanek–Zakrzewski paper is used here as a supporting theorem source; it does not explicitly identify the Hajek–Russo problem as solved.

Limitations

This packet does not prove a full positive or negative answer in ZFC. The positive theorem depends on the existence of a point-countable family of positive-measure closed sets of size ω_2 . The negative theorem depends on ω_2 being a caliber of Radon measures. The relationship between these hypotheses and standard forcing axioms was not settled in this packet.

Human Review Recommendation

The Banach-space/topological proofs above are short and should be checked first for correct use of Talagrand’s theorem and the κ -Corson theorems. The main mathematical next step is set-theoretic: determine whether the point-countable positive family at ω_2 is consistent, or whether a ZFC argument forces a contradiction without assuming the measure-caliber hypothesis.

References

- [1] P. Hajek and T. Russo, *An uncountable version of Ptak's combinatorial lemma*, J. Math. Anal. Appl. 470 (2019), 1070–1080; arXiv:1805.06028.
- [2] W. Marciszewski, G. Plebanek, and K. Zakrzewski, *Digging into the classes of κ -Corson compact spaces*, arXiv:2107.02513.
- [3] M. Talagrand, *Sur les espaces de Banach contenant $\ell_1(\tau)$* , Israel J. Math. 40 (1981), 324–330.